

Requirements Table

Telemedicine Slot Booking & Prescription Portal

Actors: Patient, Attending Physician, Payment Gateway (external system).

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Set
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures prescriptions are legally valid, tamper-evident, and traceable to a licensed physician.	Set 4
FR-002	Functional	The system shall allow a patient to search and filter available physicians by medical specialty and real-time slot availability.	High	Pass: Results show only physicians with ≥ 1 open slot in the next 7 days, correctly filtered by specialty. Fail: Results include unavailable or wrong-specialty physicians.	Core discovery function patients need before booking; reduces failed booking attempts.	Set 1
FR-003	Functional	The system shall allow a patient to book a specific video consultation slot, reserving it exclusively for that patient once payment is confirmed.	High	Pass: The booked slot disappears from all other patients' availability lists within 2 seconds. Fail: The same slot can be booked by two different patients.	Prevents double-booking and guarantees the integrity of the appointment schedule.	Sets 1, 2
FR-004	Functional	The system shall automatically generate and transmit a unique, encrypted video-consultation room link to both patient and physician no later than 10 minutes before the scheduled appointment.	High	Pass: Both parties receive a working, single-use encrypted link ≥ 10 minutes before start time. Fail: Link is missing, reused across sessions, or delivered late.	Ensures secure, timely, and private access to the tele-consultation room.	Set 3
FR-005	Functional	The system shall allow a patient to download their digitally signed prescription as a PDF from their account within 24 hours of consultation completion.	Medium	Pass: Patient downloads a valid, signed PDF from the dashboard within 24 hours of consultation end. Fail: PDF is unavailable, unsigned, or late.	Gives patients timely access to prescriptions required for pharmacy fulfillment.	Set 4
NFR-001	Nonfunctional (Performance & Security)	All video-session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking confirms target latency and security standards under simulated peak load. Fail: Any data found unencrypted in transit or at rest.	Protects sensitive patient health data (PHI) and satisfies healthcare data-protection obligations.	All sets
NFR-002	Nonfunctional (Performance & Scalability)	The system shall support at least 500 concurrent video-consultation sessions with end-to-end video latency under 300 ms.	High	Pass: Load test at 500 concurrent sessions shows average latency < 300 ms with zero session drops. Fail: Latency exceeds 300 ms or sessions fail under peak load.	Guarantees usable, real-time video quality during peak clinic hours.	Set 3
NFR-003	Nonfunctional (Availability & Reliability)	The system shall maintain at least 99.9% uptime during published clinic hours, with no confirmed booking or signed prescription lost during a failover.	High	Pass: 30-day monitoring shows ≤ 43 min downtime and a failover drill loses zero records. Fail: Uptime below 99.9% or any record lost.	Missed consultations and lost prescriptions carry direct clinical risk.	All sets

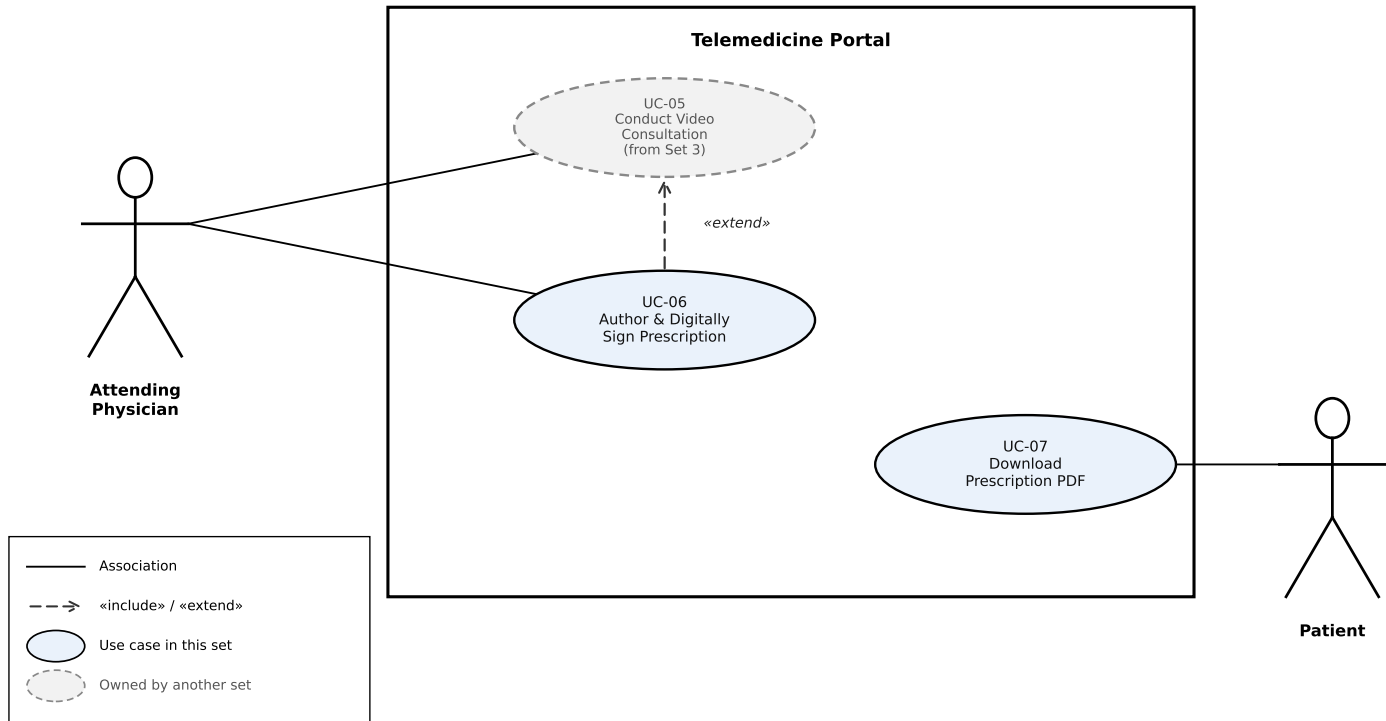
Module Split & Weightage

The eight use cases are divided into four equally weighted sets. Each set has its own use-case diagram and its own one-page use-case flow specification.

Set	Module Name	Use Cases Owned	Relationship Modelled	Weightage
Set 1	Doctor Discovery & Slot Booking	UC-01 Search Doctor by Specialty; UC-02 Book Consultation Slot	UC-02 «include» UC-03	25%
Set 2	Payment & Booking Changes	UC-03 Make Payment; UC-08 Cancel / Reschedule Slot	UC-02 «include» UC-03; UC-08 «extend» UC-02	25%
Set 3	Tele-Consultation Session	UC-04 Generate Encrypted Video Link; UC-05 Conduct Video Consultation	UC-05 «include» UC-04	25%
Set 4	Prescription Lifecycle	UC-06 Author & Digitally Sign Prescription; UC-07 Download Prescription PDF	UC-06 «extend» UC-05	25%

Set 4 — Prescription Lifecycle

Weightage: 25% | Use cases: UC-06, UC-07



Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Set 4 — Prescription Lifecycle | Weightage: 25%

Use Case: UC-06 — Author & Digitally Sign Prescription (*extends UC-05: Conduct Video Consultation, owned by Set 3*)

Primary Actor

Attending Physician

Preconditions

- A consultation for this patient has been completed and marked complete by the system.
- The physician is authenticated and holds a valid, unexpired medical licence on file.
- The physician's digital signing certificate is active and not revoked.

Postconditions

- Success: A cryptographically signed prescription PDF is stored against the patient record and released for download (UC-07).
- Failure: No prescription is issued; the draft is retained and the failure reason is written to the audit log.

Main Success Scenario

1. Physician opens the completed consultation record and selects “Create Prescription”.
2. System displays a structured prescription form pre-filled with patient and consultation details.
3. Physician enters each drug with dosage, frequency, duration, and any special instructions.
4. System validates the entries against the drug formulary and flags dosage or interaction warnings.
5. Physician reviews the warnings, adjusts the entries if required, and submits the prescription.
6. System re-validates the physician's licence status and signing certificate.
7. System applies the physician's digital signature and compiles the prescription into a PDF.
8. System stores the signed PDF against the patient record, writes an audit entry, and notifies the patient that it is available for download.
9. Use case ends successfully.

Alternate Flow — 6a. Licence or Signing Certificate Invalid

- a. 6a1. System finds the physician's licence expired or the signing certificate revoked.
- b. 6a2. System blocks signing, saves the prescription as an unsigned draft, and displays the specific reason to the physician.
- c. 6a3. System writes the blocked attempt to the audit log and alerts the clinic administrator; no PDF is generated and the patient is not notified.